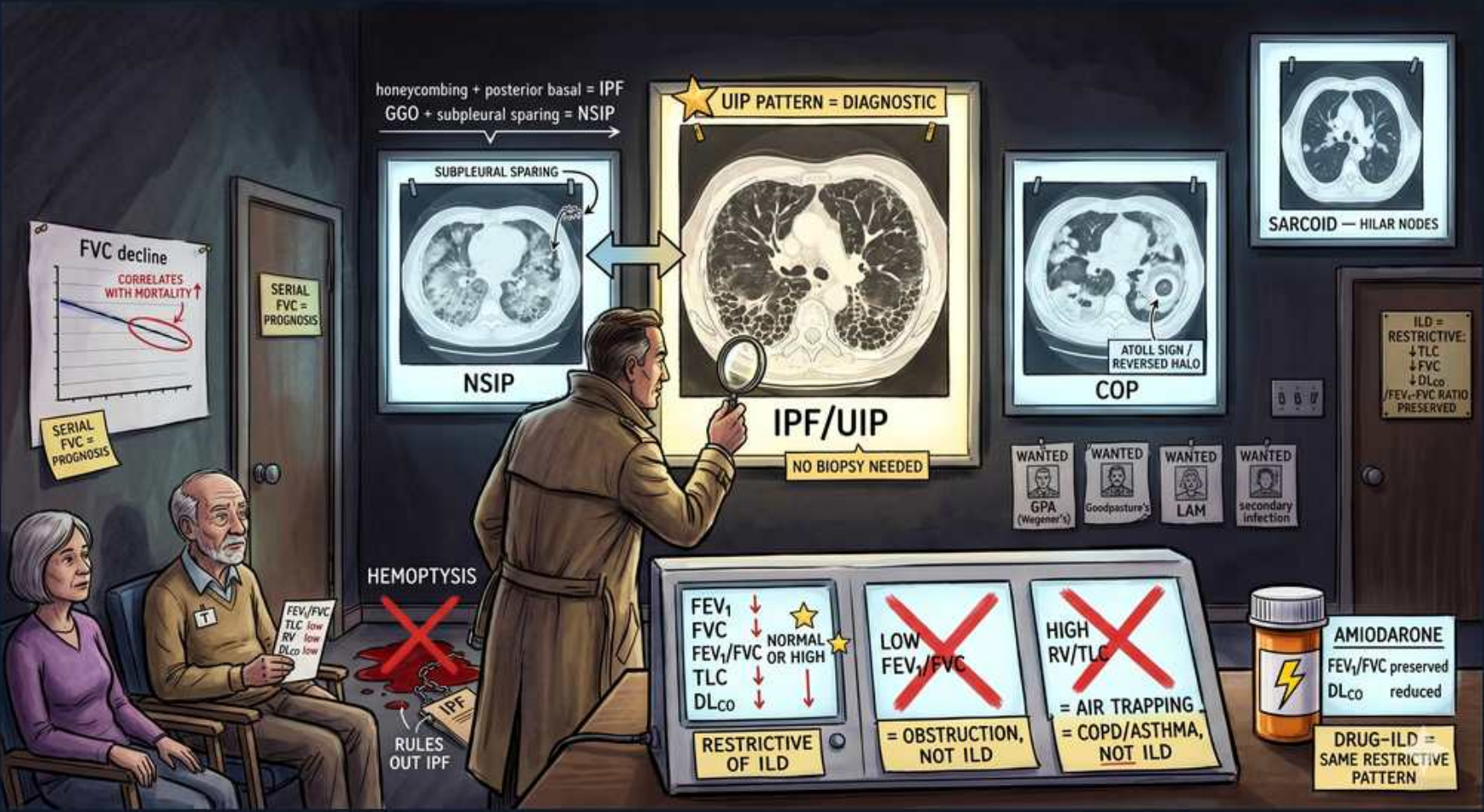


ILD — Diagnosis & Imaging Patterns (UIP/NSIP/COP) + PFTs



1. FVC decline = mortality

WHAT YOU SEE

Wall chart of falling FVC curve labeled 'correlates with mortality'

WHAT IT ENCODES

Serial FVC is THE prognostic and monitoring marker in IPF/fibrotic ILD. An absolute decline in FVC of >=10% predicted over 6–12 months marks disease progression and predicts higher mortality, and is a key trigger for escalating therapy or transplant referral. A 5–10% decline plus worsening symptoms also counts as progression.

BOARD TRAP

WRONG answer: 'follow a single baseline DLCO to gauge prognosis.' Discriminator — a one-time value is far less useful than the TRAJECTORY; serial FVC trend over 6–12 months is the validated prognostic and progression metric.

2. Serial FVC = prognosis

WHAT YOU SEE

Second note reading 'serial FVC = prognosis'

WHAT IT ENCODES

Pulmonary function should be measured SERIALLY (typically every 3–6 months) so the slope of decline guides management. A flat FVC suggests stability; a falling FVC defines a progressive fibrosing phenotype and supports antifibrotic therapy even in non-IPF ILD (per the INBUILD framework).

BOARD TRAP

WRONG answer: 'reassure a stable-feeling patient and stop monitoring PFTs.' Discriminator — symptoms lag function; only serial spirometry detects subclinical progression, so discontinuing surveillance is the error even when the patient feels unchanged.

3. Elderly couple seated

WHAT YOU SEE

An older grey-haired man and woman sitting together in the waiting-room chairs — the elderly seated pair IS the typical IPF demographic (over 60, dry cough, Velcro crackles).

WHAT IT ENCODES

IPF is a disease of older adults (typically >60), with male and ever-smoker predominance, presenting with chronic dry cough, exertional dyspnea, bibasilar 'Velcro' crackles, and clubbing. This demographic + a UIP HRCT pattern lets you diagnose IPF without biopsy.

BOARD TRAP

WRONG answer: diagnosing IPF in a young patient. Discriminator — youth, female sex, or extrapulmonary features (Raynaud, arthritis, rash, sicca) point to CTD-ILD, NSIP, or sarcoidosis; serologies (ANA, RF, anti-CCP, myositis panel) are mandatory before calling ILD 'idiopathic.'

4. Hemoptysis = NOT IPF

WHAT YOU SEE

Card 'hemoptysis' with a red X, 'rules out IPF'

WHAT IT ENCODES

Hemoptysis is NOT a feature of IPF and should redirect the differential: diffuse alveolar hemorrhage from ANCA vasculitis (GPA/MPA) or anti-GBM, infection (TB, aspergillus), pulmonary embolism with infarct, or malignancy. The red X is teaching you that blood in the sputum argues against IPF.

BOARD TRAP

WRONG answer: 'attribute hemoptysis to the patient's known IPF.' Discriminator — IPF does not bleed; new hemoptysis demands a workup for vasculitis/DAH (ANCA, anti-GBM, urinalysis for glomerulonephritis), infection, or cancer.

5. NSIP scan (GGO, sparing)

WHAT YOU SEE

CT panel labeled NSIP with ground-glass and 'subpleural sparing'

WHAT IT ENCODES

NSIP shows bilateral, symmetric, lower-zone GROUND-GLASS opacities with characteristic SUBPLEURAL SPARING and little/no honeycombing. It is frequently associated with CTD (scleroderma, polymyositis/anti-synthetase, RA) and is the steroid-responsive ILD with a better prognosis than UIP. Cellular NSIP responds best; fibrotic NSIP less so.

BOARD TRAP

WRONG answer: 'subpleural honeycombing means NSIP.' Discriminator — honeycombing is the UIP hallmark; NSIP's signature is ground-glass with subpleural SPARING. Misreading the pattern flips both prognosis and treatment (immunosuppression vs antifibrotic).

6. UIP/IPF central scan

WHAT YOU SEE

Large central CT under magnifier labeled IPF/UIP, 'UIP pattern = diagnostic'

WHAT IT ENCODES

A DEFINITE UIP pattern on HRCT — subpleural, basal-predominant reticulation with HONEYCOMBING and traction bronchiectasis, no features inconsistent with UIP — establishes IPF in the right clinical context (older patient, no CTD/exposure) without surgical biopsy. This is the central, most-tested teaching point of the scene.

BOARD TRAP

WRONG answer: 'biopsy is required to confirm IPF even with definite UIP on HRCT.' Discriminator — definite UIP pattern + compatible clinical picture is diagnostic; biopsy is reserved for indeterminate/probable patterns or atypical features.

7. Honeycombing/basal = IPF

WHAT YOU SEE

Top text 'honeycombing + posterior basal = IPF'

WHAT IT ENCODES

UIP geography is SUBPLEURAL + BASAL (posterior lower lobes) predominant, with clustered cystic 'honeycombing' (stacked thick-walled cysts) and traction bronchiectasis. Memory line: 'bottom + back + honeycomb = IPF.' This distribution distinguishes UIP from the upper-lobe diseases.

BOARD TRAP

WRONG answer: 'upper-lobe predominant fibrosis = IPF.' Discriminator — UPPER-lobe predominance points away from IPF toward chronic hypersensitivity pneumonitis, sarcoidosis, silicosis, or coal worker's pneumoconiosis (mnemonic for upper-lobe: 'CASSET / FASSTEN' patterns), not UIP.

8. No biopsy needed

WHAT YOU SEE

Banner 'no biopsy needed' under the IPF scan

WHAT IT ENCODES

When HRCT shows a DEFINITE UIP pattern in an appropriate clinical setting, surgical lung biopsy is unnecessary and avoids its morbidity/mortality. Multidisciplinary discussion (pulmonology + radiology + pathology) is the standard for confirming the diagnosis when patterns are less certain.

BOARD TRAP

WRONG answer: 'order a surgical lung biopsy first' in classic UIP. Discriminator — biopsy is indicated for indeterminate/probable UIP or atypical features (to exclude HP, NSIP, sarcoid), not for definite UIP, where it adds risk without changing the diagnosis.

9. COP / atoll (reversed halo)

WHAT YOU SEE

CT panel labeled COP with 'atoll sign / reversed halo'

WHAT IT ENCODES

Cryptogenic organizing pneumonia presents subacutely (cough, dyspnea, fevers, often post-infectious or 'failed pneumonia') with patchy, often peripheral or peribronchial consolidations and the REVERSED-HALO / ATOLL sign (central ground-glass ringed by consolidation). It is exquisitely steroid-responsive with a good prognosis.

BOARD TRAP

WRONG answer: 'treat COP with antifibrotics' or 'call the atoll sign IPF.' Discriminator — COP is inflammatory and responds dramatically to corticosteroids; the atoll/reversed-halo sign and migratory consolidations distinguish it from the fibrotic, antifibrotic-treated UIP.

10. Sarcoid = hilar nodes

WHAT YOU SEE

Scan labeled 'sarcoid = hilar nodes'

WHAT IT ENCODES

Sarcoidosis: bilateral symmetric HILAR lymphadenopathy with upper-lobe-predominant perilymphatic micronodularity; biopsy shows non-caseating granulomas. Often younger patients, may have erythema nodosum, uveitis, hypercalcemia (1-alpha-hydroxylase), and elevated ACE. First-line therapy when needed is corticosteroids.

BOARD TRAP

WRONG answer: 'bilateral hilar adenopathy with upper-lobe disease = IPF.' Discriminator — hilar lymphadenopathy is a sarcoid (or lymphoma/infection) finding, not UIP; IPF is basal/peripheral without prominent adenopathy.

11. Restrictive PFTs (low TLC/FVC, high ratio)

WHAT YOU SEE

Console: normal/high FEV1/FVC, low FVC & TLC & DLCO, star 'restrictive ILD'

WHAT IT ENCODES

ILD produces a RESTRICTIVE pattern: reduced TLC and FVC, a PRESERVED or INCREASED FEV1/FVC ratio (often >0.80 because both numerator and denominator fall, denominator more), and a low DLCO. TLC is the defining measure of restriction. Reduced compliance ('stiff lungs') is the mechanism.

BOARD TRAP

WRONG answer: 'a reduced FEV1/FVC ratio confirms ILD.' Discriminator — a LOW ratio means OBSTRUCTION (COPD/asthma); restrictive ILD shows a normal/high ratio with low TLC. Confusing the ratio direction is the classic PFT trap.

12. Low DLCO

WHAT YOU SEE

A red downward DLCO arrow lit up on the PFT console — the falling DLCO IS the earliest gas-transfer defect in ILD, often dropping before spirometric restriction appears.

WHAT IT ENCODES

A reduced DLCO reflects impaired alveolar–capillary gas transfer and is often the EARLIEST PFT abnormality in ILD, sometimes before spirometric restriction appears. A disproportionately low DLCO relative to FVC (low FVC/DLCO ratio) suggests coexisting pulmonary hypertension or emphysema (combined pulmonary fibrosis and emphysema).

BOARD TRAP

WRONG answer: 'a low DLCO is specific for ILD.' Discriminator — low DLCO is nonspecific (also emphysema, pulmonary vascular disease, anemia); it must be interpreted with lung volumes — restrictive volumes + low DLCO point to ILD, hyperinflation + low DLCO points to emphysema.

13. High RV/TLC = air trapping

WHAT YOU SEE

Panel with red X: 'high RV/TLC = air trapping = COPD/asthma, not ILD'

WHAT IT ENCODES

An elevated RV and RV/TLC ratio indicates AIR TRAPPING/hyperinflation, the hallmark of obstructive disease (COPD, asthma) — the opposite of restriction, where volumes are uniformly small. The red X means air trapping argues against a pure restrictive ILD.

BOARD TRAP

WRONG answer: 'air trapping with hyperinflation is consistent with classic ILD.' Discriminator — increased RV/TLC = obstruction; ILD has decreased TLC and RV. The exception worth knowing is combined pulmonary fibrosis and emphysema, where volumes can look deceptively normal but DLCO is markedly low.

14. Amiodarone drug-ILD

WHAT YOU SEE

Pill bottle labeled amiodarone, 'drug-ILD = same restrictive pattern'

WHAT IT ENCODES

Amiodarone pulmonary toxicity causes drug-induced ILD with the same restrictive PFTs and low DLCO; a falling DLCO is the most sensitive early sign, so monitor it. Other key drug causes: methotrexate, nitrofurantoin (acute hypersensitivity or chronic fibrosis), bleomycin, and immune checkpoint inhibitors. Always scrub the medication list.

BOARD TRAP

WRONG answer: 'label restrictive ILD idiopathic without reviewing medications.' Discriminator — drug-induced ILD is reversible if the offending agent is stopped early; missing amiodarone/methotrexate/nitrofurantoin/bleomycin on the med list is the high-yield error, since management differs entirely from IPF.